Q&A

Things You Wanted to Know!

Q1. What is micro:bit board? What are the main differences between micro:bit and BeagleBone?

Talib Mirza

A1. micro:bit board is an open source embedded system having a display consisting of 25 LEDs. It was launched by BBC in 2015 for use in computer education in the UK. It is also referred to as BBC micro:bit. The main goal of micro:bit is to give educators and parents an easy-to-use tool to teach the basics of computer programming. It has been reported that the board is now available to schools, clubs and families across the US and Canada.

You can code BBC micro:bit using Blocks, JavaScript and Python. To get started with BBC micro:bit, please visit *https://microbit.org/guide/quick/*

Fig. 1: BBC micro:bit (Credit: https://en.wikipedia.org)

Major Differences Between BBC micro:bit and BeagleBone Black

Parameters	BBC micro:bit	BBB
Size	Half credit card	Credit card
Processor	ARM Cortex-M0	ARM Cortex-A8
Sensors	Accelerometer, magnetometer	No
USB	Micro	Type A, mini
Network	No	Ethernet
Bluetooth	Yes	No
Power supply	5V USB or battery	5V USB or adaptor
Manufacturer	BBC	Texas Instruments

Fig. 2: BeagleBone Black (Credit: https://en.wikipedia.org)

There are many differences between BBC micro:bit and BeagleBone Black (BBB). BBB is much more powerful than micro:bit. Some differences are listed in the table.

BeagleBone Black. This is a low-power open source single-board computer manufactured by Texas Instruments. The board was developed as an educational board that could be used in colleges around the world to teach open source hardware and software capabilities. It is also sold to the public under Creative Commons share-alike licence.

Q2. What are magnetic switches? Please give examples of their practical applications.

Pratima

A2. A magnetic switch is an electrical switch that makes or breaks contacts in the presence of a magnetic field. That is, the switch remains actuated as long as a strong magnetic field is present and opens when the field is removed. Examples of magnetic switches include reed switches, Hall effect sensors and electromechanical relays.

Some examples of applications are:

Reed switch. It is used to detect opening of a door, or as proximity switch for a security alarm.

Hall effect sensor. It is used to measure magnitude of the magnetic field. Its output voltage is directly proportional to the magnetic field strength through it. It is commonly used in internal combustion engine ignition timing, tachometers and anti-lock braking systems.

Electromechanical relay. The traditional or electromechanical relay uses an electromagnet to close or open contacts.

Answers compiled by Sani Theo, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: **editsec@efy.in**) and should include name and address of the sender.